CHAPTER 11

The d and f Block Elements

Sr. No.	Concept	Description	Other information
1.	**General electronic configuration of d - Block elements**	$(n-1)d^{1-10}ns^{1-2}$ Exceptions: $Cr = 4s^1 3d^5$ $Cu = 4s^1 3d^{10}$ $Pd = 5s^0 4d^{10}$	
2.	**Transition series:** (a) 1st Transition series or 3d series (b) 2nd Transition series or 4d series (c) 3rd Transition series or 5d series (d) 4th Transition series or 6d series	 $_{21}Sc - _{30}Zn$ $_{39}Y - _{48}Cd$ $_{57}La$, $_{72}Hf - _{80}Hg$ $_{89}Ac$, $_{104}Rf - _{112}Cn$	
3.	**Oxidation state of d - Block elements**	Common oxidation state = +2, +3	Other oxidation state = +5,+6,+7, +8
4.	**Physical properties of d - Block elements** (a) Atomic radii (b) Metallic radii (M)/Ionic radii (c) Ionization enthalpy (d) Melting point (e) Density	 In 3d series $Sc > Ti > V > Cr > Mn \geq Fe \simeq Co \simeq Ni \leq Cu < Zn$ In a group $3d < 4d \simeq 5d$ $Sc > Ti > Mn \simeq Zn > V > Cr > Fe > Cu > Co \simeq Ni$ $Zn > Fe > Cu > Co > Ni > Mn > Ti > Cr > V > Sc$ $Sc < Ti < V < Cr > Mn < Fe > Co > Ni > Cu > Zn$ $Sc < Ti < V < Cr < Mn < Zn < Fe < Co \leq Ni \simeq Cu$ In a group $3d < 4d << 5d$	

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<tr><td>5.</td><td>Sum important properties of d-block elements
(a) Standard electrode potential E^0/V
(i) M^{2+} / M
(ii) M^{3+} / M^{2+}</td><td>

Ti < V < Cr > Mn > Fe < Co < Ni < Cu > Zn
Ti < V > Cr < Mn > Fe < Co < Ni < Cu > Zn</td><td></td></tr>
<tr><td></td><td>(b) Magnetic properties
(i) Paramagnetic substances</td><td>Weakly attracted by magnetic field.</td><td>Unpaired electrons are present in paramagnetic species.</td></tr>
<tr><td></td><td>(ii) Dimagnetic substances</td><td>Weakly repelled by magnetic field.</td><td>Dimagnetic has 0 unpaired electrons.</td></tr>
<tr><td></td><td>(iii) Magnetic moment</td><td>The ‘spin only’ magnetic moment can be calculated from the relation:
$\mu = \sqrt{n(n+2)}$ B.M.</td><td>n = no. of unpaired electrons.
μ = Magnetic moment
B.M. Bohr magneton</td></tr>
<tr><td></td><td>(iv) Ferromagnetic substances</td><td>Attracted very strongly by magnetic field.</td><td></td></tr>
<tr><td></td><td>(c) Formation of Coloured Ions</td><td>
<table>
<tr><th>Configuration</th><th>Example</th><th>Colour</th></tr>
<tr><td>$3d^0$</td><td>Sc^{3+}</td><td>colourless</td></tr>
<tr><td>$3d^0$</td><td>Ti^{4+}</td><td>colourless</td></tr>
<tr><td>$3d^1$</td><td>Ti^{3+}</td><td>purple</td></tr>
<tr><td>$3d^1$</td><td>V^{4+}</td><td>blue</td></tr>
<tr><td>$3d^2$</td><td>V^{3+}</td><td>green</td></tr>
<tr><td>$3d^3$</td><td>V^{2+}</td><td>violet</td></tr>
<tr><td>$3d^3$</td><td>Cr^{3+}</td><td>violet</td></tr>
<tr><td>$3d^4$</td><td>Mn^{3+}</td><td>violet</td></tr>
<tr><td>$3d^4$</td><td>Cr^{2+}</td><td>blue</td></tr>
<tr><td>$3d^5$</td><td>Mn^{2+}</td><td>pink</td></tr>
<tr><td>$3d^5$</td><td>Fe^{3+}</td><td>yellow</td></tr>
<tr><td>$3d^6$</td><td>Fe^{2+}</td><td>Green</td></tr>
<tr><td>$3d^6 3d^7$</td><td>Co^{3+} Co^{2+}</td><td>bluepink</td></tr>
<tr><td>$3d^8$</td><td>Ni^{2+}</td><td>green</td></tr>
<tr><td>$3d^9$</td><td>Cu^{2+}</td><td>blue</td></tr>
<tr><td>$3d^{10}$</td><td>Zn^{2+}</td><td>colourless</td></tr>
</table>
</td><td></td></tr>
<tr><td></td><td>(d) Formation of Complex Compounds</td><td>Metal ions bind a number of anions or neutral molecules giving complex species e.g.
$[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$,
$[Cu(NH_3)_4]^{2+}$ and $[PtCl_4]^{2-}$</td><td></td></tr>
</table>

Sr. No.	Concept	Description	Other information
	(e) Catalytic Properties	Transition metals & their compounds are known for their catalytic activity. Ability to adopt multiple oxidation states and to form complexes. **Catalysts** Contact process = V_2O_5 Haber process = Fe Decomposition of $KClO_3$ = MnO_2 Ostwald process = Pt/Rh Zeigler Natta catalyst = $TiCl_4 + (C_2H_5)_3Al$ Hydrogenation of Alkene = Ni/Pd Wilkinson's catalyst = $RhCl(PPh_3)_3$	
	(f) Interstitial compounds	Non - stoichiometric and are neither ionic nor covalent.	e.g. TiC
	(g) Alloy formation	Due to similar atomic sizes	
6.	**Compounds of d-block elements** (a) Potassium dichromate (b) Potassium permanganate	 $K_2Cr_2O_7$ $KMnO_4$	
7.	**General electronic configuration of f - Block elements**	$(n-2)\ f^{1-14}(n-1)d^{0-1}ns^2$	
8.	**Lanthanoids** (i) Electronic configuration (ii) Oxidation states (iii) Atomic and ionic sizes	 $4f^{1-14}5d^{0-1}6s^2$(Gd: $4f^7 5d^1 6s^2$) Most common is +3 Decreases from La to Lu (Eu is the largest)	 Other O.S. = +2, +4
9.	**Actinoids** (i) Electronic configuration (ii) Oxidation states (iii) Ionic size	 $[Rn]5f^{1-14}6d^{0-1}7s^2$ Most common is +3 Gradual decrease along the series	 Other O.S. = +4, +5, +6, +7